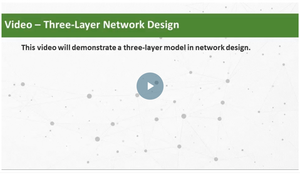
Hierarchical Networks

11.1.1

Video - Three-Layer Network Design



11.1.2

The Need to Scale the Network

Our digital world is changing. The ability to access the Internet and the corporate network is no longer confined to physical offices, geographical locations, or time zones. In today's globalized workplace, employees can access resources from anywhere in the world and information must be available at any time, and on any device. These requirements drive the need to build next-generation networks that are secure, reliable, and highly available.

These next-generation networks must not only support current expectations and equipment but must also be able to integrate legacy platforms. Businesses increasingly rely on their network infrastructure to provide mission-critical services. As businesses grow and evolve, they hire more employees, open branch offices, and expand into global markets. These changes directly affect the requirements of a network which must be able to scale to meet the needs of business.

Click Play in the figure to view an animation of a small network expanding into a larger network.

A network must support the exchange of various types of network traffic, including data files, email, IP telephony, and video applications for multiple business units. All enterprise networks must be able to do the following:

- Support critical applications
- Support converged network traffic
- Support diverse business needs
- Provide centralized administrative control

The LAN is the networking infrastructure that provides access to network communication services and resources for end users and devices. The end users and devices may be spread over a single floor or building. You create a campus network by interconnecting a group of LANs that are spread over a small geographic area. Campus network designs include small networks that use a single LAN switch, up to very large networks with thousands of connections.

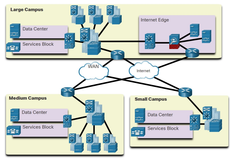
11.1.3

Borderless Switched Networks

With the increasing demands of the converged network, the network must be developed with an architectural approach that embeds intelligence, simplifies operations, and is scalable to meet future demands. One of the more recent developments in network design is the Cisco Borderless Network.

The Cisco Borderless Network is a network architecture that combines innovation and design. It allows organizations to support a borderless network that can connect anyone, anywhere, anytime, on any device, securely, reliably, and seamlessly. This architecture is designed to address IT and business challenges, such as supporting the converged network and changing work patterns.

The Cisco Borderless Network provides the framework to unify wired and wireless access, including policy, access control, and performance management across many different device types. Using this architecture, the borderless network, shown in the figure, is built on a hierarchical infrastructure of hardware that is scalable and resilient.



By combining this hardware infrastructure with policy-based software solutions, the Cisco Borderless Network provides two primary sets of services: network services, and user and endpoint services under the umbrella of an integrated management solution. It enables different network elements to work together, and allows users to access resources from any place, at any time, while providing optimization, scalability, and security.

11.1.4

Hierarchy in the Borderless Switched Network

Creating a borderless switched network requires that sound network design principles are used to ensure maximum availability, flexibility, security, and manageability. The borderless switched network must deliver on current requirements and future required services and technologies. Borderless switched network design guidelines are built upon the following principles:

- **Hierarchical** – The design facilitates understanding the role of each device at every tier, simplifies deployment, operation, and management, and reduces fault domains at every tier.
- **Modularity** – The design allows seamless network expansion and integrated service enablement on an on-demand basis.
- **Resiliency** – The design satisfies user expectations for keeping the network always on.
- **Flexibility** – The design allows intelligent traffic load sharing by using all network resources.

These are not independent principles. Understanding how each principle fits in the context of the others is critical. Designing a borderless switched network in a hierarchical fashion creates a foundation that allows network designers to overlay security, mobility, and unified communication features. Two time-tested and proven hierarchical design frameworks for campus networks are the three-tier layer and the two-tier layer model.

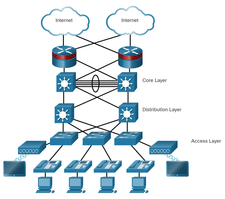
The three critical layers within these tiered designs are the access, distribution, and core layers. Each layer can be seen as a well-defined, structured module with specific roles and functions in the campus network. Introducing modularity into the campus hierarchical design further ensures that the campus network remains resilient and flexible enough to provide critical network services. Modularity also helps to allow for growth and changes that occur over time.

Click each button for an example of each design.

Three-Tier Model

Two-Tier Model

Three-Tier Model



11.1.5

Access, Distribution, and Core Layer Functions

The access, distribution, and core layers perform specific functions in a hierarchical network design.

Click each button for a description of the functions of each layer.

Access Layer

Distribution Layer

Core Layer

Access Layer

The access layer represents the network edge, where traffic enters or exits the campus network. Traditionally, the primary function of an access layer switch is to provide network access to the user. Access layer switches connect to distribution layer switches, which implement network foundation technologies such as routing, quality of service, and security.

To meet network application and end user demand, the next-generation switching platforms now provide more converged, integrated, and intelligent services to various types of endpoints at the network edge. Building intelligence into access layer switches allows applications to operate on the network more efficiently and securely.

11.1.6

Three-Tier and Two-Tier Examples

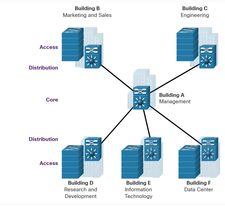
Click each button for an example and explanation of a three-tier and two-tier design.

Three-Tier Example

Two-Tier Example

Three-Tier Example

The figure shows a three-tier campus network design for organizations where the access, distribution, and core are each separate layers. To build a simplified, scalable, cost-effective, and efficient physical cable layout design, the recommendation is to build an extended-star physical network topology from a centralized building location to all other buildings on the same campus.



11.1.7

Role of Switched Networks

The role of switched networks has evolved dramatically in the last two decades. It was not long ago that flat Layer 2 switched networks were the norm. Flat Layer 2 switched networks relied on the Ethernet and the widespread use of hub repeaters to propagate LAN traffic throughout an organization.

As shown in the figure, networks have fundamentally changed to switched LANs in a hierarchical network.

A switched LAN allows additional flexibility, traffic management, quality of service, and security. It also affords support for wireless networking and connectivity, and support for other technologies such as IP telephone and mobility services.

11.1.8

Check Your Understanding - Hierarchical Networks

Check your understanding of hierarchical networks by choosing the BEST answer to the following questions.

1. Which term is used to describe a network that is always accessible?

- Hierarchical
- Modular
- Resilient
- Flexible

2. Which characteristic of a network allows it to expand and provide on-demand services?

- Hierarchical
- Modularity
- Resilient
- Flexible

3. Which term describes the ability of a network to use all available network resources and to provide data load sharing?

- Hierarchical
- Modular
- Resilient
- Flexible

4. Which network layer provides fault isolation and high-speed backbone switch connectivity?

- Core
- Distribution
- Access

5. Which network layer provides direct, switched network connectivity to the user?

- Core
- Distribution
- Access

6. Which network layer integrates with the backbone and users to provide intelligent switching, routing, and security?

- Core
- Distribution
- Access

Check

Show Me

Reset